

User Manual — LLM Evaluation Interface

***Note:** This manual covers how to *use* the interface. For installation and setup, see README.md.*

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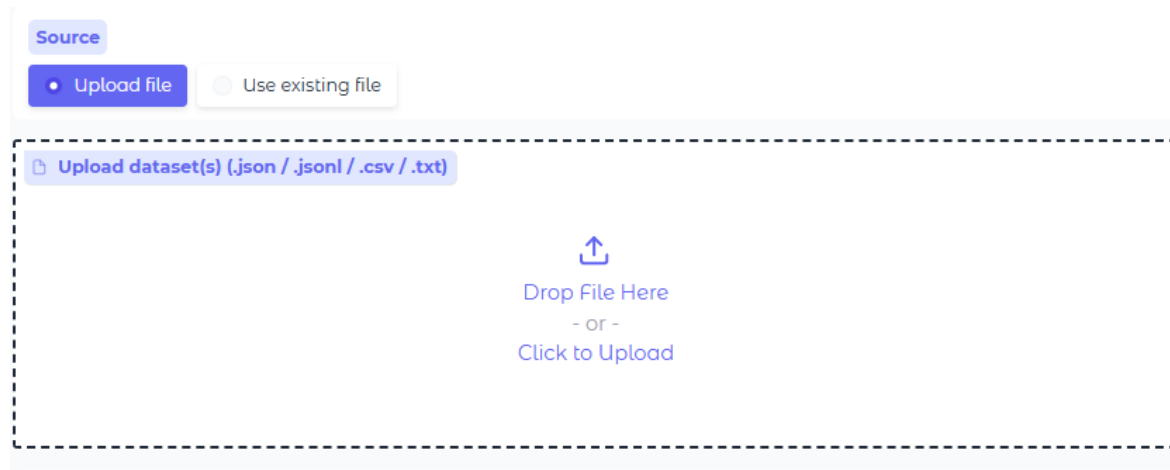
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1. Evaluation Tab

The main tab for running model evaluations on a QA dataset.

1.1 Dataset Selection



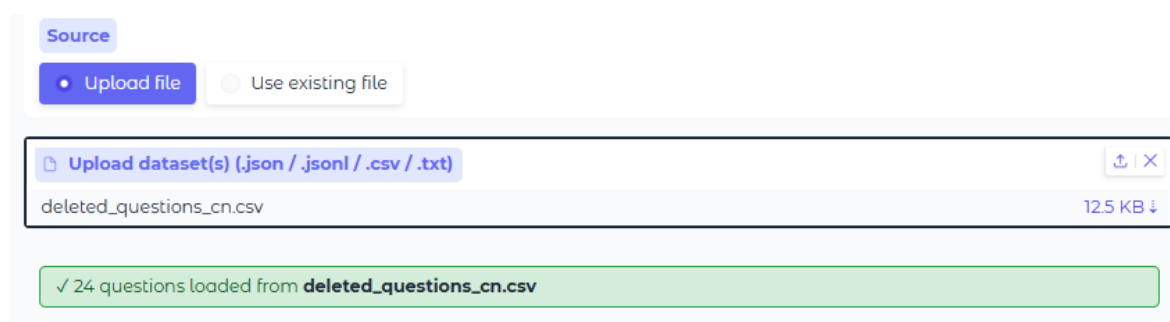
Choose how to provide your dataset:

Option	When to use
Upload file	You have a local file to upload (.json, .jsonl, .csv, .txt)
Use existing file	You have already placed a file in the <code>data/</code> folder

After loading a dataset, a **status indicator** appears:

Color	Meaning
Green ✓	Dataset is valid and ready to use
Yellow ⚠	Dataset loaded but has warnings (e.g. missing reference answers)
Red ✗	File could not be parsed or has critical errors

SUCCESS UPLOAD:



If you upload multiple files at once, a dropdown will appear — select which file to use for evaluation.

✓ deleted_questions_cn.csv

eval_results (3).csv

eval_results (2).csv

eval_results (1).csv

deleted_questions_cn.csv

1.2 Model Selection

Models

Select models to evaluate

☒ Qwen2-7B

☐ Qwen2.5-14B

☐ Mistral-7B

☐ GPT-4o

☐ Gemini-2.5-Pro

Select one or more models to evaluate. Models are loaded from `config/models.yaml`.

Available models by default:

Model	Provider
Qwen2-7B	Fireworks AI
Qwen2.5-14B	Fireworks AI
Mistral-7B	Fireworks AI
GPT-4o	OpenAI
Gemini-2.5-Pro	Google

You can select multiple models to compare them side by side in the results table.

If you need to evaluate a model that is not included in the predefined list, you can manually add it to the configuration file.

Please navigate to: `interface/config/models.yaml`

In this file, each model is defined using a structured configuration. To add a new model, you need to create a new entry following the existing format. For example:

```
name: Qwen2-7B
type: fireworks
model_id: accounts/fireworks/models/qwen2-7b-instruct
api_key: "${FIREWORKS_API_KEY}"
```

Notes

1. Make sure the model you add is already deployed and accessible via the specified provider.
2. Ensure your API key is correctly set in your environment variables.
3. After updating the configuration file, restart the application if required to load the new model.

1.3 Input Mode

Choose between **Text** and **Audio** mode.

- **Text** — The model receives the question as plain text.
- **Audio** — The model receives audio file. Additional configuration appears below (see 1.4).

1.4 Audio Configuration **(Audio mode only)**

⚠ Qwen2-7B do not support audio input directly. Whisper transcription is enabled — audio will be transcribed to text before inference.
 Note: Whisper outputs Traditional Chinese for Mandarin audio. Use the Script Converter tool if you need Simplified.

Audio Dataset
 Upload a **metadata CSV** (columns: audio_file, reference_answer) and the matching **audio files** below.

☒ Enable Whisper transcription for non-audio models (Qwen, Mistral → Whisper → text → model)

⚠ **Chinese audio note:** Whisper outputs **Traditional Chinese** for Mandarin audio. Use the Script Converter in the Tools tab to convert to Simplified Chinese.

Audio metadata CSV

Drop File Here
- or -
Click to Upload

Audio files (.wav / .mp3 / .m4a)

Drop File Here
- or -
Click to Upload

This section is only visible when **Audio** input mode is selected. Besides, if the model you selected does not support audio input, you will see the warning below.

Whisper Transcription

Setting	Behavior
Enabled	Audio is first transcribed by OpenAI Whisper. Non-audio models (e.g. Qwen, Mistral) receive the transcription as text input.
Disabled	Non-audio models are skipped . Only GPT-4o and Gemini receive audio natively.

***Note for Mandarin audio:** Whisper transcribes Mandarin in Traditional Chinese. Use the **Script Converter** tool (Tools tab) to convert to Simplified Chinese before evaluation.*

Audio Metadata CSV

Upload a CSV file with the following columns:

Column	Required	Description
audio_file	Yes	Filename of the audio file (e.g. question_01.wav)
reference_answer	Yes	The expected correct answer

Audio Files

Upload the actual audio files (.wav, .mp3, .m4a, .ogg, .flac, .webm). The filenames must match the audio_file column in your CSV.

1.5 Generation Settings

Setting	Default	Description
Max Tokens	512	Maximum number of tokens the model can generate per response
Temperature	0.0	Randomness of output. 0 = deterministic, higher = more varied
Question Limit	*(blank = all)*	Only evaluate the first N rows. Useful for quick tests.
Skip BERTScore	Checked	BERTScore requires PyTorch and is slow. Uncheck only if you need it.

1.6 Running Evaluation & Reading Results

Log

Dataset : data/sample_qa.json (5 questions)
Mode : Text
Models : ['Gemini-2.5-Pro']
Metrics : BLEU · METEOR · Token-F1 · ROUGE (BERTScore skipped)

Loading Gemini-2.5-Pro..
[Gemini-2.5-Pro] question 1/5 (4.8s)
[Gemini-2.5-Pro] question 2/5 (18.2s) ⚠ Truncated at max_tokens (finish_reason=MAX_TOKENS) — increase max_tokens in config
[Gemini-2.5-Pro] question 3/5 (5.8s)
[Gemini-2.5-Pro] question 4/5 (14.1s)
[Gemini-2.5-Pro] question 5/5 (19.9s) ⚠ Truncated at max_tokens (finish_reason=MAX_TOKENS) — increase max_tokens in config
Gemini-2.5-Pro unloaded.

✅ Evaluation complete!

Click **Run Evaluation** to start. The **Progress Log** updates in real time.

The **Results Table** shows a per-model summary (mean scores across all questions):

Column	Description
Model	Model name
ROUGE-1 F / ROUGE-2 F / ROUGE-L F	Text overlap scores (see Glossary)
BLEU	Translation-style precision score
METEOR	Recall-weighted overlap score
Token-F1	SQuAD-style token overlap F1
BERTScore-F1	Semantic similarity score (optional)
Avg Length (w)	Average response word count
Avg Latency (s)	Average response time in seconds
SNR (dB)	*(Audio mode only)* Signal-to-noise ratio of input audio
Speech Ratio	*(Audio mode only)* Fraction of audio classified as speech
Clarity Score	*(Audio mode only)* Combined audio quality score
Whisper Conf.	*(Audio mode only)* Whisper transcription confidence

1.7 Downloading Results

After evaluation completes, click **Download CSV** to save the full per-question results to `eval_results.csv`.

2. Noise Robustness Tab

Tests how well models maintain performance when audio quality degrades.

Evaluation

Noise Robustness

Tools

Noise Robustness Evaluation

Tests how model accuracy degrades as audio quality decreases.

How it works:

1. Upload the same audio dataset used in the Evaluation tab.
2. Select noise levels — the system automatically adds calibrated white noise at each SNR level (Clean + 20 dB + 10 dB + 5 dB).
3. Runs full evaluation at every noise level and reports metric degradation.

Metric to watch: BLEU and Token-F1 should decrease as SNR drops. A model that degrades slowly is more **noise-robust**.

Audio Dataset

Audio metadata CSV (same format as Evaluation tab)

Drop File Here

- Or -

Click to Upload

Audio files (.wav / .mp3 / .m4a)

Drop File Here

- Or -

Click to Upload

Models & Settings

Models to evaluate

☒ Quen2-7B

☐ Quen2.5-14B

☐ Mistral-7B

☐ GPT-4o

☐ Gemini-2.5-Pro

Noise levels to test

☒ Clean

☒ SNR 20 dB

☒ SNR 10 dB

☒ SNR 5 dB

☒ Enable Whisper transcription for non-audio models

Max tokens

512

Temperature

0

Run Noise Robustness Test

Progress

2.1 What is SNR?

Signal-to-Noise Ratio (SNR) measures how much louder the speech is compared to background noise. Lower SNR = noisier audio.

SNR Level	Audio Quality
Clean	No noise added
20 dB	Slight background noise
10 dB	Noticeable background noise
5 dB	Heavy noise, speech is hard to hear

2.2 Setup

Upload the same audio dataset (CSV + audio files) as you would in the Evaluation tab.

2.3 Noise Level Selection

Select which SNR levels to test. All four are selected by default. Uncheck any level to skip it.

2.4 Running the Test

Click **Run Noise Robustness Test**. The system will:

- For each selected noise level, add calibrated white noise to your audio files
- Run a full evaluation at that noise level
- Combine all results into a single **pivot table**


```

--- Testing noise level: Clean ---
[Clean] Loading Gemini-2.5-Pro...
[Clean] [Gemini-2.5-Pro] question 1/6 (10.1s)
[Clean] [Gemini-2.5-Pro] question 2/6 (9.4s)
[Clean] [Gemini-2.5-Pro] question 3/6 (6.3s)
[Clean] [Gemini-2.5-Pro] question 4/6 (6.3s)
[Clean] [Gemini-2.5-Pro] question 5/6 (12.3s)
[Clean] [Gemini-2.5-Pro] question 6/6 (4.0s)
[Clean] Gemini-2.5-Pro unloaded.

--- Testing noise level: SNR 20 dB ---
[SNR 20 dB] Loading Gemini-2.5-Pro...

```

The pivot table shows metrics (BLEU, Token-F1, ROUGE-1, Latency) for each model at each noise level. A model that degrades slowly across noise levels is considered **noise-robust**.

Mean BLEU / Token-F1 per model x noise level

Model	Noise Level	BLEU	Token-F1	ROUGE-1	Latency(s)
Gemini-2.5-Pro	Clean	0.0636	0.2406	0.2967	8.0418
Gemini-2.5-Pro	SNR 10 dB	0.0494	0.2252	0.2991	8.5151
Gemini-2.5-Pro	SNR 20 dB	0.0464	0.2288	0.2809	8.2594
Gemini-2.5-Pro	SNR 5 dB	0.0464	0.2461	0.2946	6.9135

Fig: Results

2.5 Downloading Results

Click **Download CSV** to save the full noise robustness results to `noise_robustness_results.csv`.

3. Tools Tab

Three utility sub-tabs for managing configuration and data quality.

3.1 Model Config

Model Configuration Editor

Edit model names, types, IDs, and API keys below. Click **Save Config** to write changes to `config/models.yaml`. The model list in the Evaluation tab will refresh automatically.

Models (edit cells directly)

name	type	model_id	api_key
Qwen2-7B	fireworks	accounts/fireworks/models/qwen2-7b-instruct	\${FIREWORKS_API_KEY}
Qwen2.5-14B	fireworks	accounts/fireworks/models/qwen2p5-14b-instruct	\${FIREWORKS_API_KEY}
Mistral-7B	fireworks	accounts/fireworks/models/mistral-7b-instruct	\${FIREWORKS_API_KEY}
GPT-4o	openai	gpt-4o-audio-preview	\${OPENAI_API_KEY}
Gemini-2.5-Pro	gemini	gemini-2.5-pro	\${GOOGLE_API_KEY}

Add New Model

Save Config **Reload from File**

View, edit, and add models to `config/models.yaml`.

Editing existing models: Click any cell in the table to edit it directly.

Adding a new model:

1. Expand the **Add New Model** section
2. Fill in:

- **Name** — display name (e.g. GPT-4o-turbo)
- **Type** — one of: openai, gemini, fireworks, hf_local
- **Model ID** — the provider's model identifier (e.g. gpt-4-turbo)
- **API Key** — use `${ENV_VAR_NAME}` to reference an environment variable

3. Click **+ Add Model**
4. Click **Save Config** to write changes to disk

After saving, the new model immediately appears in the Evaluation tab's model list.

3.2 Script Converter(For Chinese dataset only)

Chinese Script Converter

Upload a dataset file (.json / .jsonl / .csv / .txt) to convert **all text fields** between Traditional and Simplified Chinese. The converted file can be downloaded and used directly in the Evaluation tab.

Useful when Whisper transcribes Mandarin audio as Traditional Chinese — convert to Simplified before evaluation.

[Upload Dataset](#)


Drop File Here
- or -
[Click to Upload](#)

Traditional → Simplified

Simplified → Traditional

Status

[Download Converted Dataset](#)

Converts Chinese text between Traditional and Simplified characters.

When to use: OpenAI Whisper transcribes Mandarin audio in Traditional Chinese. If your reference answers are in Simplified Chinese, convert the Whisper output before evaluation to avoid scoring mismatches.

Steps:

1. Upload your dataset file (.json, .jsonl, .csv, .txt)
2. Click **Traditional → Simplified** or **Simplified → Traditional**
3. Download the converted file

The output file has the same format as the input, with `_simplified` or `_traditional` appended to the filename.

3.3 Dataset Cleaner

Dataset Quality Manager

Upload a dataset to **detect issues** (garbled characters, missing fields, meaningless symbols, excess whitespace, duplicates) and optionally **auto-clean** it before evaluation.

Upload Dataset

Drop File Here
- or -
Click to Upload

Analyze

Clean & Fix

Checks performed

Issue	Detection rule
Garbled characters	Unicode replacement char '\ufffd', null bytes, control characters
Meaningless content	Non-empty but contains no alphanumeric characters
Missing question	question field is empty
Missing reference	reference_answer field is empty
Excess whitespace	Leading/trailing spaces or internal runs of 2+ spaces
Duplicate rows	Identical (question, reference) pair appearing more than once

Detects and fixes data quality problems before running an expensive evaluation.

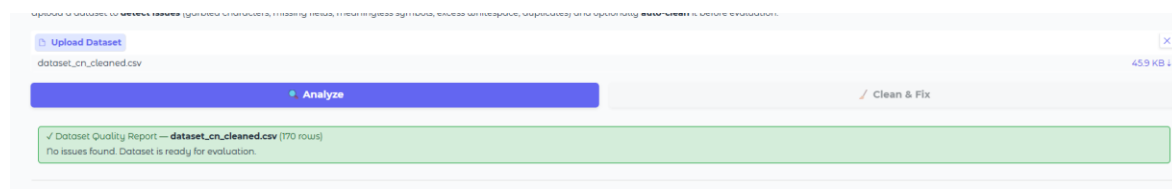
Steps:

- 1.Upload your dataset file
- 2.Click **Analyze** — the tool scans for:

Issue	Description
Garbled characters	Corrupted bytes, null characters
Meaningless content	Fields with no readable text
Missing question	Empty question field
Missing reference	Empty reference answer
Excess whitespace	Leading/trailing spaces or double spaces
Duplicate rows	Identical (question, reference) pairs

- 3.Review the status box — it shows how many rows have each issue

- 4.Click **Clean & Fix** to apply automatic repairs



The summary shows:

- **Original rows** — total rows before cleaning
- **Rows fixed** — rows that had issues and were corrected

- **Rows dropped** — rows that were removed (empty or duplicate)
- **Retained** — final row count

Dataset Quality Report — dataset_en_cleaned_simplified.txt (205 rows)		
Issue	Count	Row #s (first 10)
Garbled characters	0	—
Meaningless content	0	—
Missing question	0	—
Missing reference answer	34	21, 22, 23, 24, 25, 26, 37, 38, 39, 40, ... (+24 more)
Excess whitespace	0	—
Duplicate rows	0	—
Clean rows: 171 / 205		

Fig dataset cleaning warning

5. Click **Download** to save the cleaned dataset

4. Dataset Format Reference

Supported Formats

The interface accepts **JSON**, **JSONL**, **CSV**, and **TXT** files.

JSON

```
[
  {"question": "What is the capital of France?", "reference_answer":
  "Paris"},
  {"question": "What is 2 + 2?", "reference_answer": "4"}
]
```

JSONL (one object per line)

```
{"question": "What is the capital of France?", "reference": "Paris"}
{"prompt": "What is 2 + 2?", "answer": "4"}
```

CSV

```
question,reference_answer
What is the capital of France?,Paris
What is 2 + 2?,4
```

TXT (tab-separated)

```
What is the capital of France?      Paris
What is 2 + 2?      4
```

Column Name Aliases

The interface automatically detects column names. You do not need to rename your columns.

Field	Accepted column names
Question	question, prompt, input, text, q
Reference answer	reference_answer, reference, answer, output, expected, label, ground_truth, a
Audio file	audio_file, audio, file, filename, audio_path

All column names are matched **case-insensitively**.

5. Metrics Glossary

Text Metrics

Metric	Range	What it measures
ROUGE-1	0–1	Overlap of individual words between prediction and reference
ROUGE-2	0–1	Overlap of two-word phrases
ROUGE-L	0–1	Longest common subsequence of words
BLEU	0–1	Precision of n-gram matches (common in translation evaluation)
METEOR	0–1	Recall-weighted word overlap, accounts for synonyms via WordNet
Token-F1	0–1	SQuAD-style F1: harmonic mean of token precision and recall
BERTScore-F1	0–1	Semantic similarity using contextual embeddings (optional, requires PyTorch)

Audio Metrics **(Audio mode only)**

Metric	Range	What it measures
SNR (dB)	Higher = better	Ratio of speech volume to background noise
Speech Ratio	0–1	Proportion of the audio classified as speech
Clarity Score	0–1	Combined score: $0.5 \times \text{normalized SNR} + 0.5 \times \text{Whisper confidence}$

Whisper Confidence	0–1	Transcription confidence from Whisper (1 = fully confident)
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6. Common Workflows

Workflow A: Basic Text Evaluation

1. Go to the **Evaluation** tab
2. Under **Dataset Source**, select **Upload file** and upload your CSV or JSON dataset
3. Wait for the green ✓ status indicator
4. Select one or more **models** from the checkbox list
5. Leave **Input Mode** as **Text**
6. Adjust **Max Tokens** and **Temperature** if needed (defaults are fine for most use cases)
7. Click **Run Evaluation**
8. Watch the **Progress Log** for real-time updates
9. Review the **Results Table** — models are sorted by best metric
10. Click **Download CSV** to save results

Workflow B: Audio Evaluation with Transcription

1. Go to the **Evaluation** tab
2. Set **Input Mode** to **Audio**
3. Enable **Whisper Transcription** (so non-audio models can also be evaluated)
4. Upload your **Audio Metadata CSV** (columns: `audio_file`, `reference_answer`)
5. Upload the matching **audio files**
6. Select models (including non-audio models like Qwen or Mistral — they will receive Whisper transcriptions)
7. Click **Run Evaluation**
8. Results will include audio quality metrics (SNR, Clarity Score, etc.)

Workflow C: Noise Robustness Test

1. Go to the **Noise Robustness** tab
2. Upload your audio dataset (CSV + audio files)
3. Select models
4. Leave noise levels at default (all four selected)
5. Click **Run Noise Robustness Test**
6. The test runs 4 evaluations (one per noise level) — this takes longer than a single evaluation
7. Review the pivot table: compare how each model's scores change from Clean → SNR 5 dB
8. Click **Download CSV** to save the full results

Workflow D: Dataset Cleaning Pipeline

1. Go to **Tools** → **Dataset Cleaner**
2. Upload your raw dataset
3. Click **Analyze** and review detected issues
4. Click **Clean & Fix** and check the before/after summary
5. Download the cleaned dataset
6. *(If Mandarin audio)* Go to **Tools** → **Script Converter** and convert Traditional → Simplified
7. Use the cleaned (and converted) dataset in the **Evaluation** tab

Workflow E: Add a New Model

1. Go to **Tools** → **Model Config**
2. Expand **Add New Model**
3. Fill in Name, Type, Model ID, and API Key
4. Click **+ Add Model**, then **Save Config**
5. The new model immediately appears in the Evaluation tab